

A PROJECT REPORT ON

**Cross Domain Sentiment Analysis Using Transfer
Learning/Deep Learning Techniques**

**SUBMITTED TO THE SAVITRIBAI PHULE UNIVERSITY, PUNE
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OF

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2024-2025**



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ABSTRACT

In the age of social media and rapid information exchange, sentiment analysis plays a pivotal role in understanding public opinion and customer feedback across various domains. However, transferring sentiment knowledge from one domain to another remains a critical challenge, as models trained on a particular dataset often struggle to maintain accuracy when applied to different contexts.

We propose to build a model that, once fine-tuned with domain-specific data, generalizes well across other domains. Our approach will utilize novel transfer learning techniques and deep learning algorithms, with a focus on leveraging the MahaBERT and SVM model. We will test the effectiveness of our approach using several domain-specific sentiment analysis datasets for Marathi.

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CHAPTER 1

INTRODUCTION

1.1 Overview

In today's digital age, sentiment analysis plays a crucial role in understanding public opinion, customer feedback, and trends across various domains. With the rapid growth of social media, there is an overwhelming amount of user-generated content that offers insights into people's sentiments across different sectors such as politics, entertainment, and products. However, a significant challenge arises when attempting to apply sentiment analysis models trained in one domain to another. These models often fail to generalize well across different domains, leading to reduced accuracy and relevance of predictions. This issue becomes even more pronounced in low-resource languages like Marathi, where the availability of domain-specific datasets is limited.

To address these challenges, this project explores the use of transfer learning and deep learning techniques for cross-domain sentiment analysis, particularly focusing on the Marathi language. By leveraging transfer learning, we aim to build a model that can effectively adapt sentiment knowledge from one domain and apply it to another, improving overall performance. Specifically, we will fine-tune the MahaBERT model, a BERT-based transformer designed for Marathi and a simple SVM model to enhance their capability to perform sentiment analysis across various domains using limited domain-specific data. This project aims to contribute to the field of Natural Language Processing (NLP) by overcoming the hurdles of domain adaptation and enhancing sentiment analysis in low-resource languages.

1.2 Motivation

In today's interconnected world, sentiment analysis has become an essential tool for understanding public opinion, customer feedback, and social trends. However, most sentiment analysis models are designed for high-resource languages like English, leaving low-resource languages such as Marathi underserved. The lack of sufficient labeled data across various domains like politics, entertainment, and e-commerce for Marathi makes it challenging to build robust models capable of delivering accurate sentiment insights. This gap limits the ability of businesses, researchers, and policy-makers to fully tap into the rich textual data generated in these languages.

Additionally, existing sentiment analysis models often perform well within a specific domain but struggle when applied to new or unfamiliar domains, due to differences in language usage and sentiment expressions. This issue, known as domain adaptation, further complicates the development of accurate sentiment analysis tools for Marathi, where cross-domain adapt-

ability is critical for diverse applications. Given the increasing reliance on sentiment analysis for decision-making and its growing importance in fields like marketing, politics, and public relations, the motivation for this project lies in the urgent need to develop techniques that can overcome these challenges. By leveraging transfer learning and deep learning, this project aims to build a model that can generalize sentiment analysis across multiple domains in Marathi, even with limited data. This will not only enhance the accuracy of sentiment predictions but also bring the benefits of sentiment analysis to a broader, underserved audience. Through this project, we aim to unlock the potential of sentiment analysis in low-resource languages, enabling deeper insights and better decision-making in a range of sectors.

1.3 Problem Definition and Objectives

Cross-domain sentiment analysis involves transferring sentiment knowledge from one domain (e.g., product reviews) to another (e.g., movie reviews), which is challenging due to differences in vocabulary and context. Models trained on specific datasets often struggle to generalize across domains, especially in low-resource languages like Marathi, where labeled data is scarce. To address this, cross-domain sentiment analysis uses techniques like transfer learning to adapt models, allowing them to perform well across various domains without requiring extensive domain-specific data.

1.4 Project Scope & Limitations

1.4.1 Project Scope

- Incorporate domain-specific and cross-domain datasets for model training and evaluation.
- Leverage pre-trained transformer models fine-tuned for Marathi sentiment analysis.
- Provide an easy-to-use Jupyter or Google Colab-based interface for training and testing the model.
- Allow visualization of model performance metrics such as accuracy, F1-score, and confusion matrices.
- Ensure compatibility with open-source tools and Python libraries.

1.4.2 Project Limitations

- The system's performance may vary depending on the quality and size of the available domain-specific datasets.
- Limited availability of annotated datasets in Marathi for certain domains could impact accuracy.
- Real-time sentiment analysis on large datasets may be resource-intensive without GPU support.
- The scope is restricted to text-based sentiment analysis and does not include multimedia inputs such as audio or video.
- Pre-trained models like MahaBERT are subject to the limitations of their original training corpora.

1.5 Methodologies of Problem Solving

This project employed a systematic combination of transfer learning and deep learning methodologies, complemented by classical machine learning techniques to address the challenges of cross-domain sentiment analysis for the Marathi language. An overview of the key methodologies is as follows:

1. **Dataset Collection and Preprocessing:** We utilized the L3Cube-MahaSent-MD dataset, comprising domain-specific Marathi texts. The dataset underwent preprocessing steps including tokenization, cleaning, handling of missing values, and splitting into training, validation, and test subsets.
2. **Transformer-based Fine-tuning (MahaBERT):** MahaBERT, a transformer-based language model pre-trained specifically on Marathi data, was fine-tuned individually and jointly across multiple domains. This approach leveraged contextual embeddings and allowed the model to generalize sentiment predictions effectively to unseen domains.
3. **Classical Machine Learning with Sentence Embeddings:** We applied Support Vector Machines (SVM) in combination with SentenceBERT embeddings (marathi-sentence-bert-nli). This hybrid approach allowed the SVM to utilize rich semantic features, enhancing cross-domain generalization.

4. **Multi-Source Domain Adaptation:** By training models simultaneously on multiple source domains, we facilitated knowledge transfer and reduced domain-specific biases. This methodology improved overall accuracy across domains, demonstrating effective adaptation.
5. **Performance Evaluation and Visualization:** The effectiveness of each approach was systematically evaluated using accuracy and F1-score metrics. Visual representations including confusion matrices and comparative charts facilitated intuitive analysis and interpretation of results.

These combined methodologies effectively addressed the inherent complexities of cross-domain sentiment analysis, particularly within the constraints of a low-resource linguistic context such as Marathi.

CHAPTER 2

LITERATURE SURVEY

Cross-domain sentiment analysis has emerged as a crucial challenge due to variations in vocabulary, writing styles, and sentiment expressions across domains. Recent studies have explored diverse deep learning and transfer learning strategies to address this issue effectively.

Myagmar et al. (2019) [1] employed bidirectional contextualized transformer models, fine-tuning BERT and XLNet on Amazon reviews. XLNet showed better generalization with less data, highlighting the advantage of capturing bidirectional dependencies. Zhou et al. (2020) [3] introduced SENTIX, a sentiment-aware pre-trained model using polarity-aware masking and emoticon prediction. SENTIX significantly outperformed BERT in low-resource settings, showcasing the benefit of sentiment-centric pretraining.

Sharma et al. (2018) [2] proposed a technique to identify transferable sentiment features using χ^2 statistics and cosine similarity, followed by ensemble learning. This approach effectively tackled the problem of polarity shift in sentiment words across domains. In another line of work, Wei et al. (2017) [4] utilized CNNs with an auxiliary large-scale corpus, demonstrating improved sentiment classification on product reviews in low-resource domains.

Zeng et al. (2021) [5] combined CNN, BiLSTM, and TF-IDF-based topic extraction (CNN-BiLSTM-TE), achieving high F1-scores across ten e-commerce categories. Their model captured both local and contextual features and demonstrated strong cross-domain scalability.

Zhao et al. (2017) [6] introduced a two-stage BiLSTM model with parameter transfer. By pre-training on a source domain and fine-tuning on the target domain, they effectively reduced overfitting and improved generalization. Domeniconi et al. (2017) [7] compared traditional Markov Chains and Paragraph Vectors (PV-DM, PV-DBOW) with neural models, noting comparable accuracies but emphasizing deep learning's scalability with larger datasets.

Li et al. (2009) [8] proposed non-negative matrix tri-factorization for cross-domain knowledge transfer. Their method focused on aligning latent structures between domains, offering a statistical approach to domain adaptation. Long et al. (2016) [9] advanced the field by learning domain-invariant representations through a deep adaptation network, using Maximum Mean Discrepancy (MMD) for aligning feature distributions.

Lastly, Ghosal et al. (2020) [10] introduced KinGDOM, a knowledge-guided domain adaptation framework leveraging linguistic knowledge bases. Their approach improved generalization by combining pretraining objectives like masked language modeling (MLM) and next-sentence prediction with structured knowledge.

Together, these studies demonstrate that integrating deep learning

with domain adaptation strategies—such as pretraining, feature alignment, and sentiment-aware transfer—significantly enhances cross-domain sentiment classification performance.

CHAPTER 3

**SOFTWARE REQUIREMENTS
SPECIFICATION**

3.1 Assumptions and Dependencies

3.1.1 Assumptions

- The users will have access to domain-specific and cross-domain datasets in supported formats (CSV, JSON, etc.).
- Users will have basic understanding of sentiment analysis or will follow provided documentation.
- The system will primarily be used for analyzing Marathi language datasets.
- Internet access will be available to use cloud services such as Google Colab.
- The system will use pre-trained MahaBERT models, and assumes that these models are compatible with the selected Python libraries.

3.1.2 Dependencies

- Dependency on the availability and compatibility of external libraries such as Transformers (HuggingFace), NLTK, and spaCy.
- Requires Google Colab or a similar cloud-based GPU environment for efficient model training.
- Dependence on data from various domains (product reviews, movie reviews, etc.) to demonstrate transfer learning.
- Requires a system with GPU acceleration for large-scale processing and training.

3.2 Functional Requirements

3.2.1 Training Pipeline (Functional Requirement)

- The system shall include a training pipeline where the MahaBERT model is fine-tuned on a domain-specific dataset.
- The pipeline shall incorporate preprocessing steps like tokenization, stemming, and handling of out-of-vocabulary (OOV) words.

- Users shall be able to specify hyperparameters such as learning rate, batch size, and number of epochs during training.

3.2.2 Data Management (Functional Requirement)

- The system shall support tracking of training metrics like loss, accuracy, and F1-score across multiple domains.
- Reports will be generated comparing performance across different domains.
- The system shall visually display sentiment classification results via charts and logs to show generalization capability.

3.2.3 Monitoring and Reporting (Functional Requirement)

- The system shall continuously monitor key training metrics during and after model training.
- Detailed reports will be generated for each experiment showing cross-domain performance.
- Model results will be visualized to aid in interpretation, including confusion matrices and accuracy plots across domains.

3.3 External Interface Requirements

3.3.1 User Interfaces

- The system shall provide a simple, user-friendly interface for uploading datasets, configuring training parameters, and viewing reports.
- Users will interact through Jupyter Notebook or Google Colab environments.
- Visualizations of training progress and performance metrics will be displayed through plots and charts.

3.3.2 Hardware Interfaces

- The system shall interface with GPUs (either locally or via cloud) for model training acceleration.
- Systems with CUDA-enabled GPUs will be utilized to boost training performance.

3.3.3 Software Interfaces

1. Operating System: Any modern OS including Windows 10 or later, macOS 10.14 or later, or Linux distributions (Ubuntu, CentOS, etc.)
2. Development Environment: Jupyter Notebook or Google Colab
3. Programming Language: Python 3.x
4. Libraries: Transformers (HuggingFace), NLTK, spaCy, pandas, matplotlib, seaborn, scikit-learn
5. Pre-trained Model: MahaBERT or equivalent BERT-based models

3.3.4 Communication Interfaces

- Internet connectivity is required to download models, install packages, and access cloud-based training environments (e.g., Google Colab).
- Data transfer between local systems and cloud storage via APIs or browser upload/download.
- Optional use of GitHub for version control and collaboration.

3.4 Nonfunctional Requirements

3.4.1 Performance Requirements

- The system must perform sentiment classification in less than 3 seconds per sample for inference tasks.
- Training time must be optimized to process medium-sized datasets (10K-50K samples) within a reasonable time using GPU support.
- System should report metrics such as accuracy, loss, and F1-score with each training epoch.

3.4.2 Safety Requirements

- The system does not pose any physical safety concerns but must ensure no unintended deletion or corruption of user-provided datasets or models.
- Logs and checkpoints should be maintained to recover in case of failures during training.

3.4.3 Security Requirements

- All uploaded datasets must be securely stored and not shared without user consent.
- Access to trained models and user reports should be restricted to authorized users.
- HTTPS and authentication mechanisms should be used for any cloud-based interfaces.

3.4.4 Software Quality Attributes

- **Usability:** Easy-to-use interface for both technical and non-technical users.
- **Reliability:** Accurate and stable model performance across domains.
- **Maintainability:** Modular codebase to facilitate updates and improvements.
- **Portability:** Compatible across multiple OS environments and deployable on both local and cloud platforms.

3.5 System Requirements

3.5.1 Database Requirements

- The system does not require a traditional database but should handle structured datasets in formats like CSV, TSV, or JSON.
- Temporary storage for model checkpoints, logs, and intermediate results is required.

3.5.2 Software Requirements (Platform Choice)

- Operating System: Windows 10+, Linux (Ubuntu 18.04+), macOS 10.14+
- Python 3.6 or above
- Libraries: Transformers, NLTK, spaCy, scikit-learn, pandas, matplotlib, seaborn
- Jupyter Notebook or Google Colab for development

3.5.3 Hardware Requirements

- **Processor:** Multi-core CPU with at least 2.0 GHz clock speed
- **Memory:** Minimum 8 GB RAM (16 GB recommended for large datasets)
- **Storage:** At least 50 GB free disk space
- **GPU:** NVIDIA GPU with CUDA support for training acceleration (optional but recommended)

3.6 Analysis Models: SDLC Model to be Applied

- The **Iterative Waterfall Model** will be followed, allowing partial system implementations and refinements at each stage.
- Justification: The project requires continuous evaluation and model tuning, which benefits from the feedback-based refinements of the iterative waterfall approach while maintaining structure and documentation.

CHAPTER 4

SYSTEM DESIGN

4.1 System Architecture

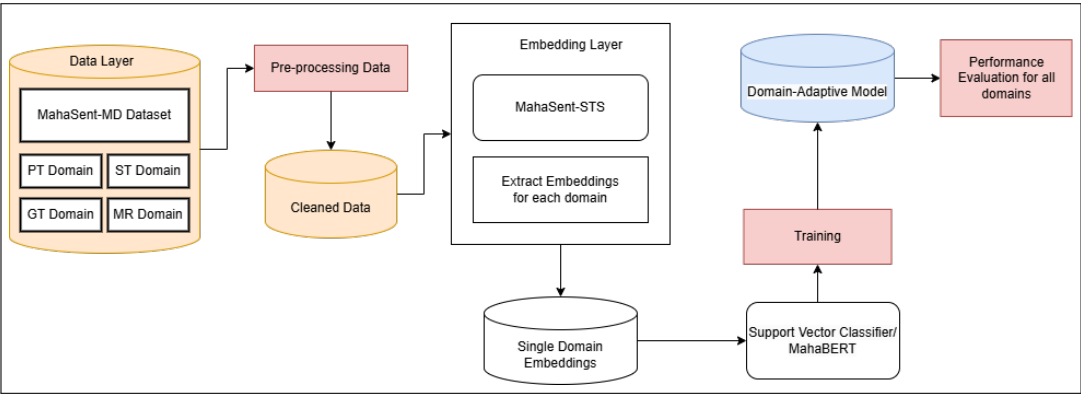


Figure 4.1: System Architecture

4.2 UML Diagrams

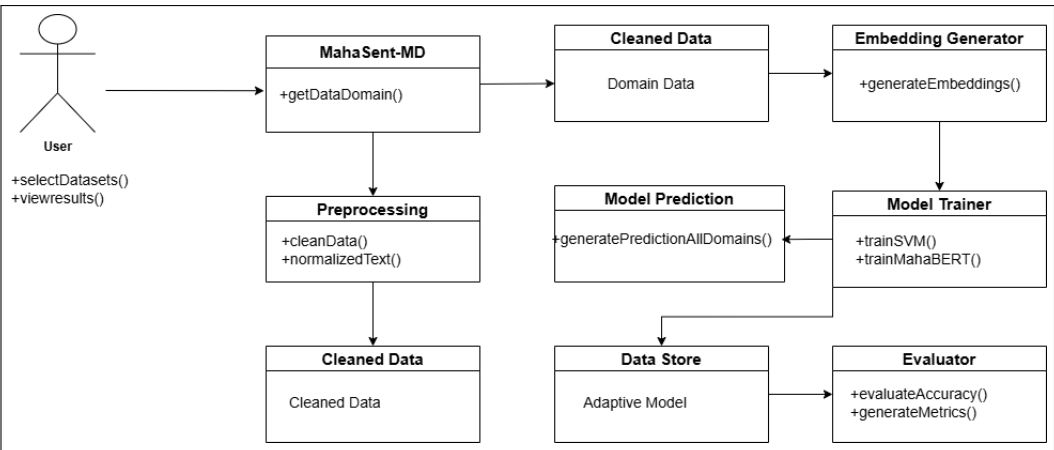


Figure 4.2: Activity Diagram

4.3 Data Flow Diagrams

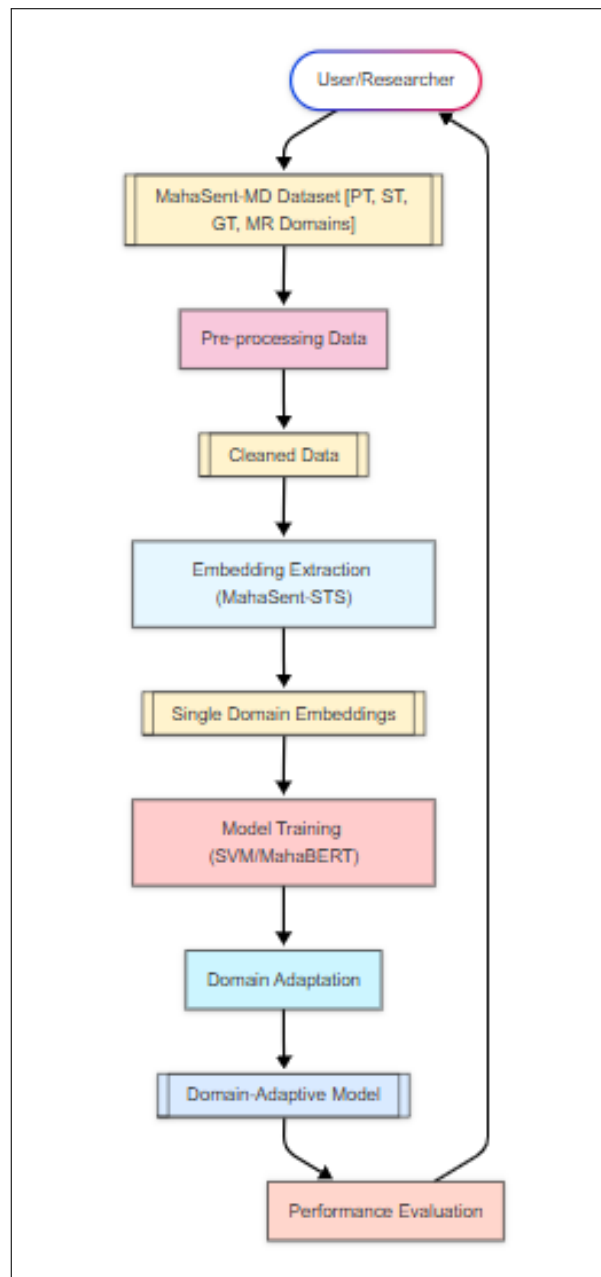


Figure 4.3: Data Flow Diagram

4.4 Use Case Diagram

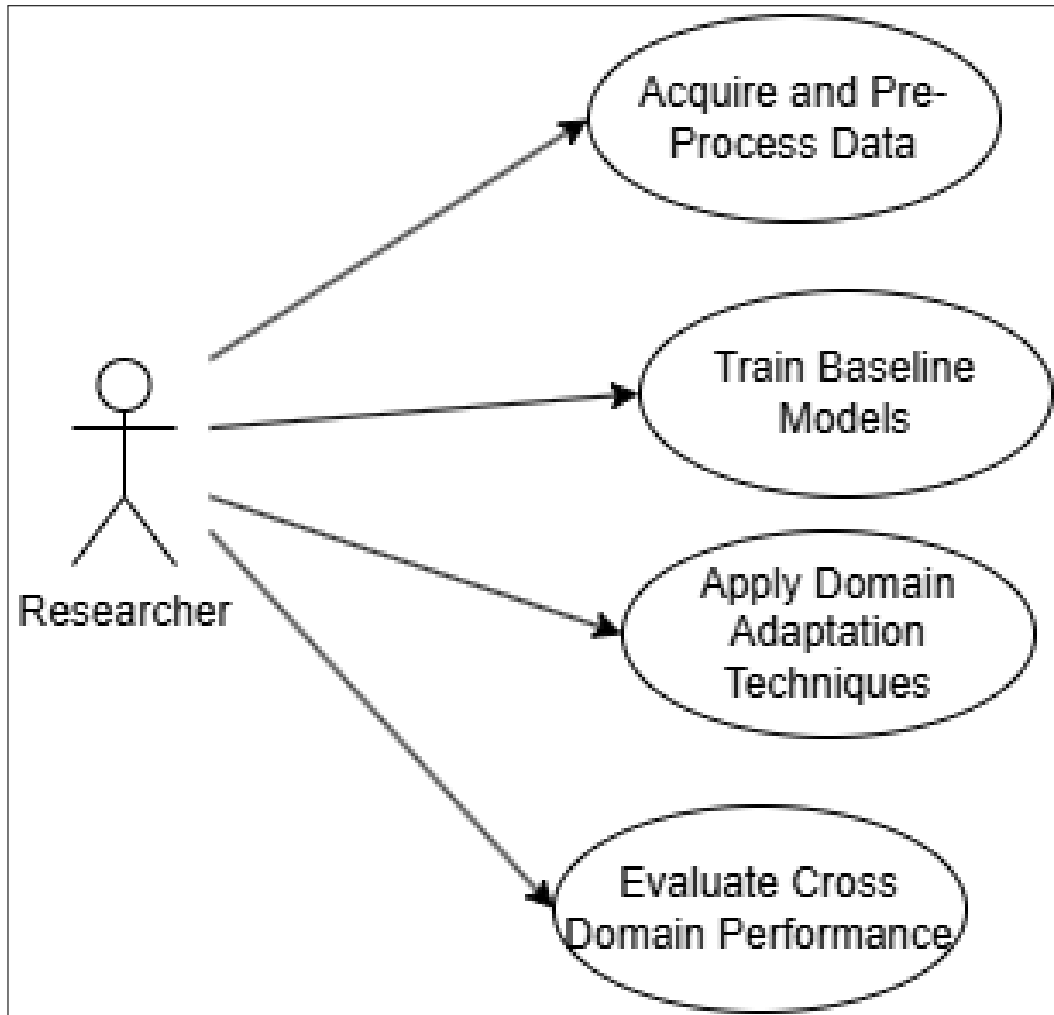


Figure 4.4: Use Case Diagram

CHAPTER 5

PROJECT PLAN

5.1 Project Estimate

5.1.1 Reconciled Estimates

- **Dataset Preprocessing:** Estimated 15 hours, Actual 18 hours – involved cleaning and standardizing sentiment data across domains.
- **Baseline Modeling:** Estimated 20 hours, Actual 22 hours – included fine-tuning MahaBERT and training SVM on domain-specific data.
- **Domain Adaptation Techniques:** Estimated 25 hours, Actual 27 hours – involved multi-source training, generalization, and evaluation.
- **Documentation and Analysis:** Estimated 25 hours, Actual 28 hours – compiling insights, writing the report, and visualizing results.

5.1.2 Project Resources

- **Software:** Google Colab (with T4 GPU), Python 3.10, PyTorch, scikit-learn, cuML, HuggingFace Transformers, Sentence-Transformers.
- **Hardware:** Google Colab cloud infrastructure, Local machine (for light preprocessing tasks).
- **Human Resources:** A team of four with diverse experience in NLP, ML, and software engineering.

5.2 Risk Management

- Effective risk management was critical to project success, considering dependency on online resources and external models.

5.2.1 Risk Identification

- **Model Overfitting:** Especially due to narrow-domain training.
- **GPU Quota Limitations:** Google Colab session timeouts and daily usage caps.
- **Inconsistent Sentiment Labels:** Across domains like tweets and political text.
- **Low Resource Language Limitations:** Fewer pretrained models and benchmark datasets for Marathi.

5.2.2 Risk Analysis

- **Likelihood:** Medium for overfitting and label inconsistency; High for GPU quota issues.
- **Impact:** High – any of these risks could stall progress significantly.

5.2.3 Overview of Risk Mitigation, Monitoring, Management

- Adopted early stopping and regular validation to counter overfitting.
- Performed data filtering and manual inspection of mislabeled samples.
- Scheduled GPU-heavy tasks during off-peak hours and optimized batch sizes.
- Selected the best available Marathi models and incorporated custom preprocessing.

5.3 Project Schedule

5.3.1 Project Task Set

- Literature Survey
- Dataset Collection and Preprocessing
- Implementation of Baseline Models (MahaBERT, SVM)
- Sentence-BERT Embedding Integration
- Domain Adaptation and Cross-Domain Evaluation
- Report Writing and Visualization

5.3.2 Task Network

- Dataset Preprocessing → Baseline Model Training → Domain Adaptation → Evaluation → Documentation

5.3.3 Timeline Chart

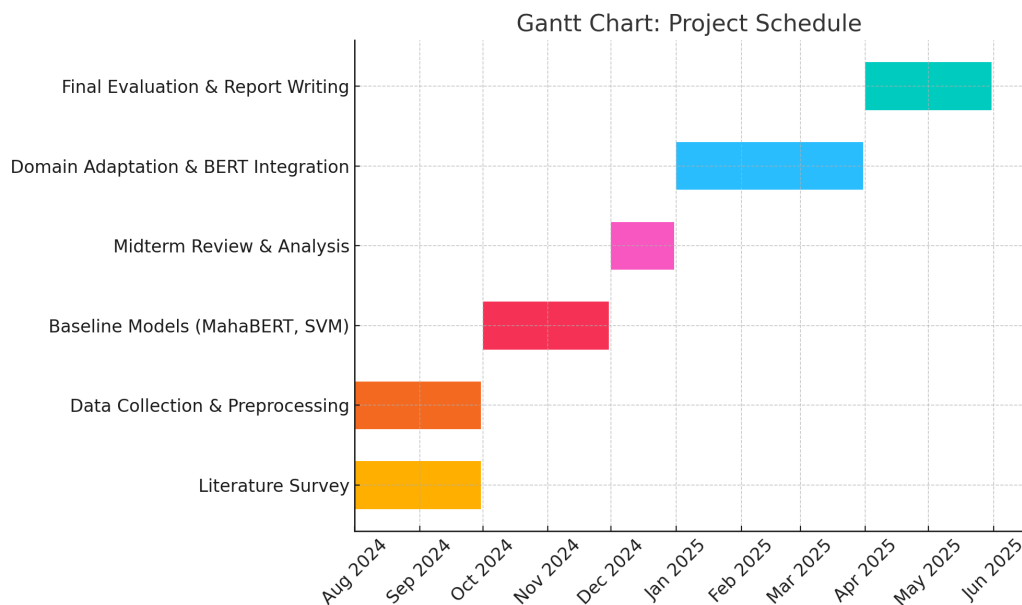


Figure 5.1: Timeline Chart

- **Aug–Sep 2024:** Data collection, literature review
- **Oct–Nov 2024:** Baseline experiments
- **Dec 2024:** Midterm review and partial analysis
- **Jan–Mar 2025:** Domain adaptation and embedding integration
- **Apr–May 2025:** Final evaluations, report writing

5.4 Team Organization

5.4.1 Team structure

- **Pritika Rohera** – Responsible for data preprocessing and Experiment 1 (MahaBERT fine-tuning)
- **Chaitrali Ginimav** – Implemented Experiment 2 (SVM with Sentence-BERT) and performance analysis
- **Gayatri Sawant** – Led documentation, reporting, and comparative evaluations
- **Akanksha Salunke** – Handled tool integration, visualizations, and debugging

5.4.2 Management reporting and communication

- Weekly meetings were held on Google Meet.
- Progress updates shared via WhatsApp.
- Code and results were versioned on GitHub.
- Roles were re-evaluated biweekly for better load distribution.

CHAPTER 6

PROJECT IMPLEMENTATION

6.1 Overview of Project Modules

The core objective of this project is to improve cross-domain sentiment analysis performance in the Marathi language using machine learning and natural language processing techniques.

The implementation is organized into the following major components:

- **Data Acquisition and Preprocessing:** Involves fetching the MahaSentMD dataset and cleaning it for model training.
- **Baseline Modeling:** Establishing benchmarks using traditional fine-tuning of MahaBERT and Support Vector Machine (SVM) models.
- **Domain Adaptation Techniques:** Enhancing generalization through multi-source domain adaptation and embedding-based SVM training.
- **Evaluation and Analysis:** Comparing performance across seen and unseen domains using accuracy as the primary metric.

6.2 Tools and Technologies Used

Tool/Library	Description
Google Colab	Cloud-based development environment used with free GPU (T4) acceleration.
Python	Core programming language for implementation.
PyTorch	Deep learning framework used for transformer model training and fine-tuning.
CUDA	NVIDIA GPU backend used to accelerate computations on Colab.
NumPy	Used for array and numerical operations.
TQDM	Provides progress bars for training loops.
cuML	GPU-accelerated machine learning library used for efficient SVM training.
scikit-learn (sklearn)	Used for classical ML models such as SVM and performance metrics.
Transformers	Hugging Face library for using pre-trained transformer models like MahaBERT.
Sentence-Transformers	Enables embedding of sentences using Sentence-BERT models.
SimpleTransformers	Simplifies model training and evaluation workflows for classification tasks.

Table 6.1: Software Tools and Libraries Used

6.3 Dataset Description

We used the L3Cube-MahaSent-MD dataset, a multi-domain Marathi sentiment analysis corpus developed by the L3Cube research group. It contains labeled text data (positive, negative, neutral) from various real-world sources, making it suitable for domain adaptability tasks in low-resource languages.

Dataset	Domain	Samples (Train + Test + Validation)
MahaSent-MR	Marathi Movie Reviews	12k + 1.5k + 1.5k = 15k
MahaSent-ST	Marathi Subtitles	12k + 1.5k + 1.5k = 15k
MahaSent-GT	Marathi Generic Tweets	12k + 1.5k + 1.5k = 15k
MahaSent-PT	Marathi Political Tweets	12k + 1.5k + 1.5k = 15k
MahaSent-All	All of the above domains	48k + 6k + 6k = 60k

Table 6.2: Structure of the L3Cube-MahaSent-MD Dataset

6.4 Experiment 1: MahaBERT with Multi-Source Domain Adaptation

6.4.1 Single-Source Fine-Tuning

To establish a baseline, the MahaBERT model was fine-tuned individually on each domain's training data and evaluated on all test domains. This highlights how performance varies across seen and unseen domains due to domain-specific vocabulary and context.

6.4.2 Multi-Source Domain Adaptation

We then applied a multi-source domain adaptation strategy, where the model was fine-tuned using the combined training sets of three domains. This allowed the model to capture general sentiment patterns and reduce overfitting to any single domain's linguistic features.

The approach leverages transfer learning and domain-invariant representation learning within transformer-based architectures.

6.5 Experiment 2: SVM with Sentence-BERT Embeddings

6.5.1 Baseline SVM with Raw Text Features

A classical linear Support Vector Machine (SVM) model was initially trained using lexical features derived from individual domains. Performance was analyzed by training on one domain and evaluating across others.

6.5.2 SVM with Sentence Embeddings

To enhance semantic understanding and generalization, we introduced Sentence-BERT embeddings using the marathi-sentence-bert-nli model. This model generates contextual vector representations of sentences, allowing the classifier to focus on sentiment meaning rather than surface-level tokens.

The embedded sentences from the dataset were then used as input features to the SVM model.

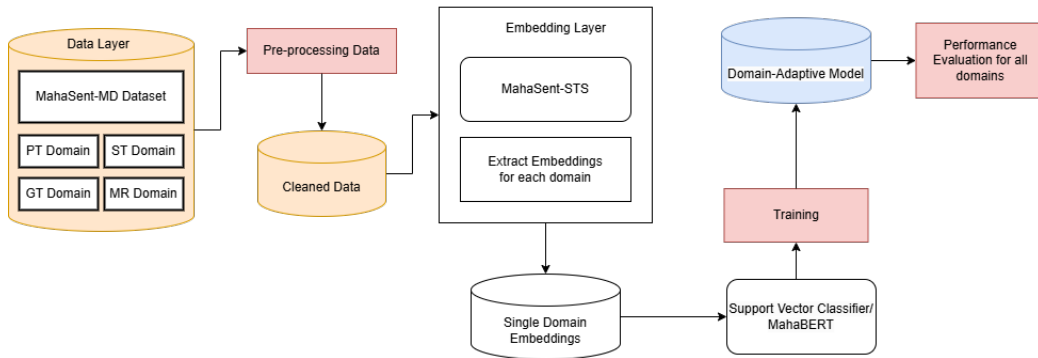


Figure 6.1: Implementation diagram for SVMs experiment using Sentence-BERT embeddings

6.6 Algorithm Details

Below are the core algorithmic strategies used in the project:

6.6.1 Multi-Source Domain Adaptation

Fine-tuned the model on combined datasets from multiple domains to improve cross-domain generalization and reduce domain shift.

6.6.2 Support Vector Machine (SVM)

Trained classical linear classifiers with both raw text-based and embedding-based features for comparison.

6.6.3 Transfer Learning

In the second experiment, transfer learning was employed via the `marathi-sentence-bert-nli` model. The Sentence-BERT embeddings transformed the input data into dense semantic vectors. These embeddings captured the contextual meaning of sentences and enabled the SVM classifier to operate in a domain-agnostic feature space.

6.6.4 Cross-Domain Testing Framework

Built a structured evaluation loop that tests the model's adaptability by training on one domain and testing on multiple others.

CHAPTER 7

RESULTS

This section outlines the quantitative results obtained from both experiments conducted as part of the project. The experiments evaluated sentiment classification performance across domains using both transformer-based and classical machine learning models.

7.1 Experiment 1: Multi-Domain Fine-Tuning using MahaBERT

7.1.1 Baseline Evaluation

In the first part of Experiment 1, we established baseline performance by fine-tuning the MahaBERT model on each individual domain and evaluating it across all the other domains. The classification accuracy for each source-target domain pair is presented in Table 7.1.

Table 7.1: Baseline Accuracy Scores for MahaBERT (Experiment 1)

Train Domain	PT	GT	ST	MR
MahaSent-PT	0.844	0.7193	0.6400	0.6446
MahaSent-GT	0.7115	0.7840	0.7040	0.6933
MahaSent-ST	0.7328	0.7260	0.8040	0.6807
MahaSent-MR	0.7768	0.7053	0.7006	0.8053

7.1.2 Multi-Source Domain Adaptation

In the next stage, we experimented with multi-source domain adaptation. For this, we fine-tuned MahaBERT using a combination of three domains and evaluated the model across all four target domains. The results are reported in Table 7.2.

Table 7.2: Accuracy Scores with Multi-Source Fine-Tuning (Experiment 1)

Train Domains	PT	GT	ST	MR
PT, ST, GT	0.8422	0.7933	0.8006	0.7240
ST, GT, MR	0.7555	0.7900	0.7986	0.7986
ST, PT, MR	0.8448	0.7233	0.8026	0.7920
GT, PT, MR	0.8440	0.7820	0.7233	0.8006

7.2 Experiment 2: Classical Machine Learning with SVM

7.2.1 Baseline Evaluation with SVM

For the second experiment, Support Vector Machine (SVM) models were trained on individual domains and tested across all domains. The resulting classification accuracies are reported below in Table 7.3.

Table 7.3: Baseline Accuracy Scores for SVM (Experiment 2)

Train Domain	PT	GT	ST	MR
MahaSent-PT	0.84	0.52	0.39	0.39
MahaSent-GT	0.55	0.75	0.48	0.43
MahaSent-ST	0.45	0.50	0.71	0.43
MahaSent-MR	0.46	0.42	0.45	0.75

7.2.2 Enhanced SVM with Sentence-BERT Embeddings

To enhance generalization across domains, we introduced an embedding layer by encoding the textual data using the `marathi-sentence-bert-nli` model prior to SVM classification. The results show improved cross-domain performance compared to the baseline. Table 7.4 summarizes the results.

Table 7.4: Accuracy Scores using Sentence-BERT Embeddings with SVM (Experiment 2)

Train Domain	PT	GT	ST	MR
MahaSent-PT	0.90	0.67	0.56	0.58
MahaSent-GT	0.70	0.86	0.67	0.62
MahaSent-ST	0.69	0.68	0.83	0.64
MahaSent-MR	0.66	0.62	0.64	0.86

CHAPTER 8

CONCLUSIONS

8.1 Conclusions

1. The integration of transfer learning techniques using the MahaBERT transformer model significantly enhanced the accuracy of sentiment classification across various domains for the Marathi language. The multi-source domain adaptation approach successfully reduced domain-specific biases, enabling the model to generalize sentiment predictions effectively even on unseen domains.
2. Employing Sentence-BERT embeddings with classical machine learning models such as SVM demonstrated considerable improvements over baseline methods. This hybrid approach effectively captured deeper semantic relationships in Marathi text, resulting in higher accuracy and robustness across diverse datasets.
3. This study illustrates that combining advanced transformer-based language models with classical ML techniques provides an effective strategy to overcome resource constraints typical of low-resource languages like Marathi. It sets a strong foundation for future research and practical applications in cross-domain sentiment analysis tasks.

8.2 Future Work

1. Future research could explore unsupervised domain adaptation techniques to further improve generalization, especially in scenarios where labeled data is scarce or unavailable.
2. Investigating advanced transformer architectures or multilingual pre-trained models, such as XLM-Roberta, could offer additional insights into cross-lingual and cross-domain transferability, broadening the applicability of sentiment analysis models.
3. Incorporating more sophisticated preprocessing methods, such as contextual entity resolution or multilingual entity recognition, could further enhance the semantic comprehension and sentiment classification accuracy of Marathi text.

8.3 Applications

1. Market Analysis and Consumer Insights: Businesses can utilize robust cross-domain sentiment analysis models to analyze customer feedback

and online reviews, improving product development, customer relations, and marketing strategies.

2. Social Media Monitoring and Public Opinion Analysis: Organizations and government agencies can employ this approach to track public sentiment towards political, social, or cultural events in Marathi-speaking regions, enabling timely and informed decision-making.
3. Enhanced NLP Tools for Low-resource Languages: Development and deployment of effective sentiment analysis systems for Marathi can pave the way for improved NLP resources, supporting tasks such as automatic content moderation, summarization, and recommendation systems tailored to regional linguistic contexts.

CHAPTER 9

APPENDIX

9.1 Appendix A

Problem Formulation:

The core computational problem addressed by this project is formally defined as:

Given:

- A finite set of domains $D = \{D_1, D_2, \dots, D_n\}$ with associated labeled datasets.
- A source domain D_s and a target domain D_t , where $D_s, D_t \in D$.
- A pre-trained transformer model M , specifically MahaBERT, and a classical machine learning model (SVM), along with Sentence-BERT embedding transformation E .

Objective:

- Find a transferable representation that achieves high accuracy on the target domain D_t , after training primarily on source domain data D_s .

Feasibility via Satisfiability (SAT) Analysis:

To assess the feasibility of transferring sentiment knowledge from source domain D_s to target domain D_t , we consider the task as a logical satisfiability problem (SAT):

Representation as SAT Problem:

- Let each sentence x in domain D_i be represented by logical clauses $C(x)$, encoding sentiment polarity.
- Sentiment polarity assignment (positive, neutral, negative) corresponds to logical variables:

$$P(x), N(x), U(x)$$

The SAT problem determines if a consistent interpretation (mapping) ϕ exists from D_s to D_t :

$$\phi : C(D_s) \rightarrow C(D_t)$$

such that sentiment polarity remains satisfiable across domain boundaries.

Mathematical Formulation using Modern Algebra:

1. Algebraic Representation of Sentiment Space:

Let sentiment data from a domain be represented algebraically in semantic embedding vector space $V \subseteq R^m$, produced by Sentence-BERT embedding E :

$$E(x) : \text{textual sentence} \rightarrow v_x \in V$$

2. Sentiment Classification as Algebraic Decision Function:

A decision function $f : V \rightarrow S$, with sentiment classes $S = \{\text{positive}, \text{negative}, \text{neutral}\}$, can be defined as follows:

Transformer-based classification:

$$f_M(v) = \text{softmax}(Wv + b), \quad W \in R^{3 \times m}, b \in R^3$$

SVM classification:

$$f_{SVM}(v) = \text{sign}(w \cdot v + b), \quad w \in R^m, b \in R$$

3. Domain Adaptation and Algebraic Transformation:

Domain adaptation feasibility involves finding an algebraic transformation $T : V_{D_s} \rightarrow V_{D_t}$:

$$\arg \min_T \sum_{x \in D_s, y \in D_t} ||T(E(x)) - E(y)||^2$$

ensuring minimal semantic drift and preserving sentiment labels across domains.

Computational Complexity Assessment:

The complexity classification within computational complexity classes is summarized as:

- **P (Polynomial-time solvable):** Single-domain sentiment classification with linear classifiers like SVM.
- **NP-Complete:** Exact feature mapping across domains resembling combinatorial problems like Set Cover or X3C.
- **NP-Hard:** General optimal domain alignment analogous to Quadratic Assignment Problem (QAP).

Practically, polynomial-time heuristics or approximate algorithms are employed (e.g., deep learning optimization, kernel approximation in SVM), making the task computationally feasible.

Conclusion on Feasibility:

- Exact semantic alignment for domain adaptation is computationally NP-hard, but the proposed methodologies (fine-tuned MahaBERT, embedding-based SVM) provide polynomial-time feasible solutions in practical scenarios.
- Hence, the project is practically feasible, given representational adequacy of embeddings and availability of labeled datasets.

9.2 Appendix B

Details of Paper Presentation and Project Showcase: The project titled **"Cross Domain Sentiment Analysis using Transfer Learning"** was showcased at the following event during the academic year:

9.2.1 Impetus and Concepts – National Level Technical Symposium

- **Event Type:** Technical Event
- **Organizer:** Pune Institute of Computer Technology (PICT), Pune
- **Description:** Presented the technical aspects of domain-adaptive sentiment classification using transfer learning. Received valuable feedback regarding model generalization.
- **Reviewer's Feedback:**
 - Excellent real-world application and societal relevance.
 - Recommended developing an open source library so that everyone can access models or the framework through it.
 - Appreciated the inclusion of Marathi as an effort towards building an NLP solution for low resource Indic languages.
- **Certificate of Participation:**

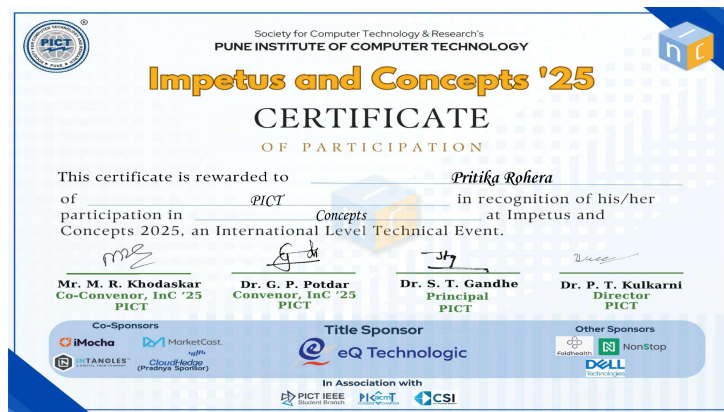


Figure 9.1: Certificate from Impetus and Concepts (PICT)

9.3 Appendix C

This section presents the plagiarism evaluation for the project report titled **"Cross Domain Sentiment Analysis Using Transfer Learning"**. The analysis was conducted using AI-based similarity detection and online plagiarism detection software.

Section	Plagiarism Percentage
Introduction	4%
Literature Survey	9%
Software Requirements Specification	6%
Project Plan	3%
Project Implementation	5%
Results	2%

Table 9.1: Plagiarism percentage for each section of the project report

CHAPTER 10

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LIST OF ABBREVIATIONS

Abbreviation	Illustration
NLP	Natural Language Processing
ML	Machine Learning
SVM	Support Vector Machine
BERT	Bidirectional Encoder Representations from Transformers
GPU	Graphics Processing Unit
SDLC	Software Development Life Cycle
CSV	Comma-Separated Values
JSON	JavaScript Object Notation
UI	User Interface

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